

Discrete Mathematics

Assignment - 08 Sept, 2026

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* Relation, Mapping, Function:

⇒ Books Reference:

- Discrete Mathematical Structures with Applications to Computer Science - J.P. Tremblay, R. Mahajan.
- Pages: 192, 193, 149

1. Give one basis / aspect / concept / foundational concept to differentiate the three terms - Relation / Mapping and Function.

A: • Chapter 2.8 - Relations and Ordering

- Relations - Any set of ordered pairs defines a binary relation. A particular ordered pair, say $(x, y) \in R$, where R is a Relation, by writing $x R y$ which may be read as " x is in relation R to y ".

• Chapter 2.4 - Functions

- Function - Let X and Y be any two sets. A relation f from X to Y is called a function if for every $x \in X$ there is a unique $y \in Y$ such that $(x, y) \in f$.
- Mapping - Terms such as "transformation", "map" or "mapping", are used as synonyms for "function". The notations $f: X \rightarrow Y$ or $X \xrightarrow{f} Y$ are used to express f as a function from X to Y .

2. Give one property of relation that is not applicable to functions and vice-versa.

A: • Chapter 2.4 - Functions

- The definition of function requires that relation must satisfy two additional conditions in order to qualify as a function. The first condition is that every $x \in X$ must be related to some $y \in Y$, that is, the domain of f must be X .

- According to this, a property of relation that is not applicable to functions is that a relation allows one input to map in its domain to not map to any output.

- "The second requirement of uniqueness can be expressed as $(x, y) \in f \wedge (x, z) \in f \Rightarrow y = z$

- We can infer from this property of function that a function must be unique, that is, a one input can only map to one unique output. A relation does not hold this property.